

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I B Munna 192372028 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

B Munna

Signature of the student:

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Assignment: Classical Encryption Techniques

Scenario 1: Event Communication Security

Aim

To encrypt and decrypt a confidential event message using the Caesar Cipher with a shift key of 4.

Theory

The Caesar Cipher is a substitution cipher in which each letter of the plaintext is shifted by a fixed number of positions in the alphabet. The same key is used for both encryption and decryption.

a. Encryption

Plaintext:

MEET AT NOON

Using a shift of +4:

Plaintext| M| E| E| T| A| T| N| O| O| N

Ciphertext| Q| I| I| X| E| X| R| S| S| R

Encrypted Mess

QIIX EX RSSR

Encryption Formula

- Encryption: $C = (P + 4) \bmod 26$

- Decryption: $P = (C - 4) \bmod 26$

Where:

- P = Plaintext
- C = Ciphertext

b. Decryption

The receiver knows the secret key (4). Each encrypted letter is shifted 4 positions backward.

Example:

- Q $\rightarrow$ M
- I $\rightarrow$ E
- X $\rightarrow$ T
- E $\rightarrow$ A

Recovered Message:
MEET AT NOON

Result

The message was successfully encrypted and decrypted using the Caesar Cipher with a shift value of 4.

Advantages

- Easy to understand and implement.
- Requires very little computation.
- Suitable for learning basic cryptography.

Limitations

- Very small key space (only 25 possible keys).
- Easily broken using brute-force attacks.
- Not secure for real-world communication.

Scenario 2: Legacy Encryption in Corporate Communication

Aim

To recover the original message from a Rail Fence Cipher and evaluate its security.
Theory

The Rail Fence Cipher is a transposition cipher. Instead of replacing letters, it rearranges their positions using a zigzag pattern across a fixed number of rails.

a. Reverse the Encryption

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

The message uses a 3-Rail Rail Fence Cipher.

Decryption Steps

1. Count the number of characters.

2. Draw a zigzag pattern with 3 rails.
3. Fill the rails with the ciphertext.
4. Read the zigzag pattern to recover the original text.

Recovered Plaintext

WEARE DISCOVERED FLEEAT ONCE

Verification

Re-encrypting the plaintext with a 3-Rail Rail Fence Cipher produces the same ciphertext, confirming that the decryption is correct.

b. Critical Evaluation

The Rail Fence Cipher is weak because:

- It only changes the order of letters.
- Letter frequencies remain unchanged.
- It has a limited number of possible rail values.
- Modern computers can recover the plaintext within seconds.
- It provides no protection against known-plaintext and brute-force attacks.
- It is unsuitable for securing confidential business information.

Security Analysis

Feature| Rail Fence

Cipher Type|

Transposition Cipher

Security Level| Very Low

Key Space| Small

Resistant to Brute Force| No

Suitable for Modern Systems| No

c. Practical Enhancement

Replace the Rail Fence Cipher with AES-256 (Advanced Encryption

Standard). Why AES-256?

- Uses a 256-bit secret key.
- Extremely difficult to crack using brute-force attacks.
- Provides confidentiality, integrity, and strong data protection.
- Used by banks, governments, and multinational companies.
- Meets modern cybersecurity standards.

Additional Security Measures

- Use strong password-based key management.
- Enable secure key exchange protocols.
- Combine encryption with digital signatures for authentication.
- Use TLS/HTTPS for secure communication over networks.

Overall Comparison

Feature| Caesar Cipher| Rail Fence Cipher|AES-256
CipherType|Substitution|Transposition| SymmetricBlock Cipher

Security| Low| Low| Very High

Key Size| 25 PossibleShifts| Few Rail Values| 256-bit Key

Vulnerable to Brute Force| Yes| Yes| Practically No

Modern Usage| Educational| Educational| Real-world Applications

Conclusion

Both the Caesar Cipher and Rail Fence Cipher are classical encryption techniques that demonstrate the basic principles of cryptography. However, they are vulnerable to brute force and cryptanalysis attacks and are not suitable for modern communication systems. Organizations should use strong encryption algorithms such as AES-256, along with secure key management and authenticated communication protocols, to ensure confidentiality, integrity, and data security.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation

				s	
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of

Results 15

Insightful

analysis with

accurate

interpretation and validation of results

Good

analysis

with

reasonable interpretation

Basic

analysis

with limited insights

Poor or incorrect analysis of results

Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation
---------------	----	---	--	--	----------------------------------